

Random variable & Mathematical Expression

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Random variable:-

A random variable is a real valued function whose values are determined with the outcomes of a random experiment. It is usually denoted by X, Y, Z .

Let us consider the experiment of tossing two fair coins. The sample space of the experiment is,

$$S = \{HH, HT, TH, TT\}$$

Let X denotes the number of heads,

So,

<u>Sample Point</u>	<u>Number of Head</u>
HH	2
HT	1
TH	1
TT	0

Here, x can take the values 0, 1, 2. Therefore, x is a random variable.

Types of Random Variable:-

1. Discrete Random variable
2. Continuous Random variable

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Types of Random Variable:-

1. Discrete Random Variable
2. Continuous Random variable

Discrete Random Variable:- A random variable is called discrete random variable if it can take only isolated values.

Ex- family members, mobile numbers etc.

Continuous Random Variable:- A random variable is called continuous random variable if it can take any values between certain limits.

Ex- Weight, Age etc.

Probability function:-

A function $f(x)$ of a discrete random variable x is called a probability function if it satisfies the following two conditions,

i. $f(x) \geq 0$

ii. $\sum f(x) = 1$

Probability Density function:-

A function $f(x)$ of a continuous random variable x is called a probability density function if it satisfies the following two conditions,

i. $f(x) \geq 0$

ii. $\int_{-\infty}^{\infty} f(x) dx = 1$

Mathematical Expectation:-

If x is a discrete or continuous random variable with probability function or probability density function $f(x)$, then the mathematical expectation of x is usually denoted by $E(x)$ or μ and defined by

$\Rightarrow \mu = E[x] = \sum x f(x)$; If x is a discrete random variable

$\Rightarrow \mu = E[x] = \int_{-\infty}^{\infty} x f(x) dx$; If x is a continuous random variable

Properties of Mathematical Expectation of a random variable:-

1. If " a " is a constant then $E[a] = a$

2. If x is a random variable with expectation $E[x]$ then $E[ax+b] =$

$$E[ax+b] = E[x] + b; \text{ where } a \text{ and } b \text{ is constant.}$$

$$3. E[ax] = aE[x]$$

4. If x and Y are random variables then

$$E[x+Y] = E[x] + E[Y]$$

5. If x and Y are random variables then

$$E[x-Y] = E[x] - E[Y]$$

☐ Properties of variance a random variable:-

1. If 'a' is a constant. then $V[a] = 0$
2. If X is a random variable, then $V[ax+b] = a^2 V[X]$
where a and b are constant.
3. $V(a, X) = a^2 V[X]$
4. $V[X+Y] = V[X] + V[Y]$
5. $V[X-Y] = V[X] - V[Y]$
6. $V[X] = E[X - E(X)]^2 = E(X)^2 - [E(X)]^2$

Prb-1
A continuous random variable X has the following probability density function,

$$f(x) = Kx^2, \quad 0 \leq x \leq 1$$

1. Determine the value of K ,
2. $P(X < 0.65)$
3. $P(X > 0.30)$
4. $P(0.25 < X < 0.75)$
5. $E[X]$ or $E[7X+9]$
6. $V[X]$ or $V[7X+9]$
7. $SD(X)$
8. $C.V(X)$

Sol:-

1. We know, $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{0.0}^{1.0} Kx^2 dx = 1$$

$$\Rightarrow K \int_{0.0}^{1.0} x^2 dx = 1$$

$$\Rightarrow K \left[\frac{x^3}{3} \right]_{0.0}^{1.0} = 1$$

$$\Rightarrow \frac{K}{3} (1^3 - 0^3) = 1$$

$$\Rightarrow K = 3$$

Therefore,

$$f(x) = 3x^2; 0 \leq x \leq 1.$$

ii) Probability that x less than 0.65 (Ans).

$$P(x < 0.65) = \int_0^{0.65} 3x^2 dx \quad [K=3]$$

$$= 3 \int_0^{0.65} x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_0^{0.65}$$

$$= (0.65)^3 - 0^3$$

$$= 0.274 \text{ (Ans).}$$

iii) Probability that x greater than 0.30 is $P[x > 0.30]$

$$P(x > 0.30) = \int_{0.30}^1 3x^2 dx \quad [K=3]$$

$$= 3 \int_{0.30}^1 x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_{0.30}^1$$

$$= 1^3 - (0.30)^3$$

$$= 0.973$$

(Ans).

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iv. Probability that lies between 0.25 and 0.75

$$P(0.25 < x < 0.75) = \int_{0.25}^{0.75} 3x^2 dx \quad [K=3]$$

$$= 3 \int_{0.25}^{0.75} x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_{0.25}^{0.75}$$

$$= (0.75)^3 - (0.25)^3$$

$$= 0.406.$$

(Ans)

$$v) E[x] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$= \int_0^1 x \cdot 3x^2 dx$$

$$= 3 \int_0^1 x^3 dx$$

$$= 3 \left[\frac{x^4}{4} \right]_0^1$$

$$= \frac{3}{4} [1^4 - 0^4]$$

$$= \frac{3}{4}.$$

$$\therefore E[7x+9] = 7 \cdot E[x] + 9$$

$$= 7 \cdot \frac{3}{4} + 9$$

$$[E[ax+b] = aE[x] + b]$$

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$$= \frac{57}{4} \cdot (\text{Ans}).$$

$$\begin{aligned} \text{vi) } V(x) &= E[x^2] - [E(x)]^2 \\ &= \int_0^1 3x^2 \cdot 3x^2 dx - \left(\frac{3}{4}\right)^2 \\ &= 3 \int_0^1 x^4 dx - \left(\frac{3}{4}\right)^2 \\ &= 3 \left[\frac{x^5}{5} \right]_0^1 - \left(\frac{3}{4}\right)^2 \\ &= \frac{3}{5} (1^5 - 0^5) - \frac{9}{16} \\ &= \frac{3}{5} - \frac{9}{16} \\ &= \frac{3}{80} \cdot (\text{Ans}). \end{aligned}$$

$$\begin{aligned} \text{vii) } (7x+9) &= 7^2 V(x) \quad [V(ax+b) = a^2 V(x)] \\ &= 49 \cdot \frac{3}{80} \\ &= \frac{147}{80} \quad (\text{Ans}). \end{aligned}$$

viii) We know,

$$\begin{aligned} SD(x) &= \sqrt{V(x)} \\ &= \sqrt{3/80} \\ &= \frac{\sqrt{15}}{20} \quad (\text{Ans}). \end{aligned}$$

viii) ~~Ans~~

We know,

$$\begin{aligned} C.V(x) &= \frac{SD(x)}{E(x)} \times 100\% \\ &= \frac{\sqrt{15}/20}{3/4} \times 100\% \\ &= 25.81\% \\ & \quad (\text{Ans}). \end{aligned}$$

Problem-2

$$f(x) = K(x+1); \quad 2 \leq x \leq 5$$

i. Find the value of K .

ii. $P(x > 3)$

iii. $P(x < 4)$

iv. $E[x]$

v. $V[x]$

vi. $SD[x]$

vii. $C.V[x]$